UNIVERSIDAD DE SAN CARLOS DE GUATEMALA
CENTRO UNIVERSITARIO DE OCCIDENTE
DEPARTAMENTO DE ESTUDIOS DE POSTGRADOS
“MAESTRIA EN INGENIERIA ESTRUCTURAL Y SISMORRESISTENTE ”
CURSO: INGENIERÍA SÍSMICA
MSc. ING. JAVIERA YOLANDA MALDONADO DE LEÓN

¨EJERCICIO ¨
REGISTRO
XN003.EVT

ESTUDIANTES:
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FECHA 19 DE AGOSTO 2026

ESPECTROS ELÁSTICOS LINEALES – REGISTRO XN003

PARÁMETROS DE LA TAREA	
Registro	XN003 (componente N-S, datos suministrados)
Intervalo de muestreo Δt	0.005000
Duración del registro (s)	60.000000
PGA del registro (gal)	75.496190
PGA del registro (g)	0.076958
Periodos evaluados	0.1 s a 4.0 s
Incremento de periodo	0.1 s
Amortiguamientos	2%, 5% y 10%
Fecha indicada en la tarea	19-08-2026, 23:59 horas

Formulas Empleadas	
Ecuación de movimiento	$\ddot{u} + 2\xi\omega_n\dot{u} + \omega_n^2 u = -\ddot{u}g(t)$
Frecuencia natural	$\omega_n = 2\pi/T$
Frecuencia amortiguada	$\omega_a = \omega_n\sqrt{(1-\xi^2)}$
Pendiente del tramo del acelerograma	$\alpha = (ug1 - ug0)/t^*$
Constante A	$A = ug0/\omega_n^2 - 2\xi\alpha/\omega_n^3$
Constante B	$B = \alpha/\omega_n^2$
Constante C	$C = u0 - A$
Constante D	$D = [\dot{u}0 + C\omega_n\xi - B]/\omega_a$
Desplazamiento espectral	$Sd = \text{máx} \lvert u(t) \rvert$
Pseudo-velocidad	$Sv = \omega_n\ Sd$
Pseudo-aceleración	$Sa = \omega_n^2\ Sd$
Conversión	1 gal = 1 cm/s²; Sa(g) = Sa(gal)/981

VALIDACIÓN	
Criterio	Se reprodujo el procedimiento del ejemplo T=0.1 s, ξ=5% y t*=0.005 s del archivo 12.1)EspectroDeRespuestaCalculo.pdf.
Resultado	Los primeros 20 pasos se calculan en la hoja Validacion_0.1s_5% con las mismas ecuaciones de la catedrática.

ENTREGABLES INCLUIDOS	
1	Tabla + gráfica del espectro para ξ=2%
2	Tabla + gráfica del espectro para ξ=5%
3	Tabla + gráfica del espectro para ξ=10%
4	Tabla + gráfica comparativa con los tres espectros

REGISTRO XN003 – DATOS DE ACELERACIÓN		
n	**t (s)**	**ag (gal = cm/s²)**
1	0.000	-0.00649
2	0.005	-0.03355
3	0.010	-0.05503
4	0.015	0.04242
5	0.020	0.37344
6	0.025	0.89455
7	0.030	1.39838
8	0.035	1.70742
9	0.040	1.86624
10	0.045	2.13676
11	0.050	2.80351
12	0.055	3.95763
13	0.060	5.40385
14	0.065	6.71389
15	0.070	7.38395
….	…..	…………….
….	…..	…………….
11982	59.905	-0.51205
11983	59.910	-0.50328
11984	59.915	-0.50193
11985	59.920	-0.50403
11986	59.925	-0.50652
11987	59.930	-0.50572
11988	59.935	-0.49604
11989	59.940	-0.47089
11990	59.945	-0.42603
11991	59.950	-0.36320
11992	59.955	-0.29029
11993	59.960	-0.21760
11994	59.965	-0.15231
11995	59.970	-0.09445
11996	59.975	-0.03649
11997	59.980	0.03291
11998	59.985	0.12424
11999	59.990	0.24385
12000	59.995	0.39149
12001	60.000	1.02407

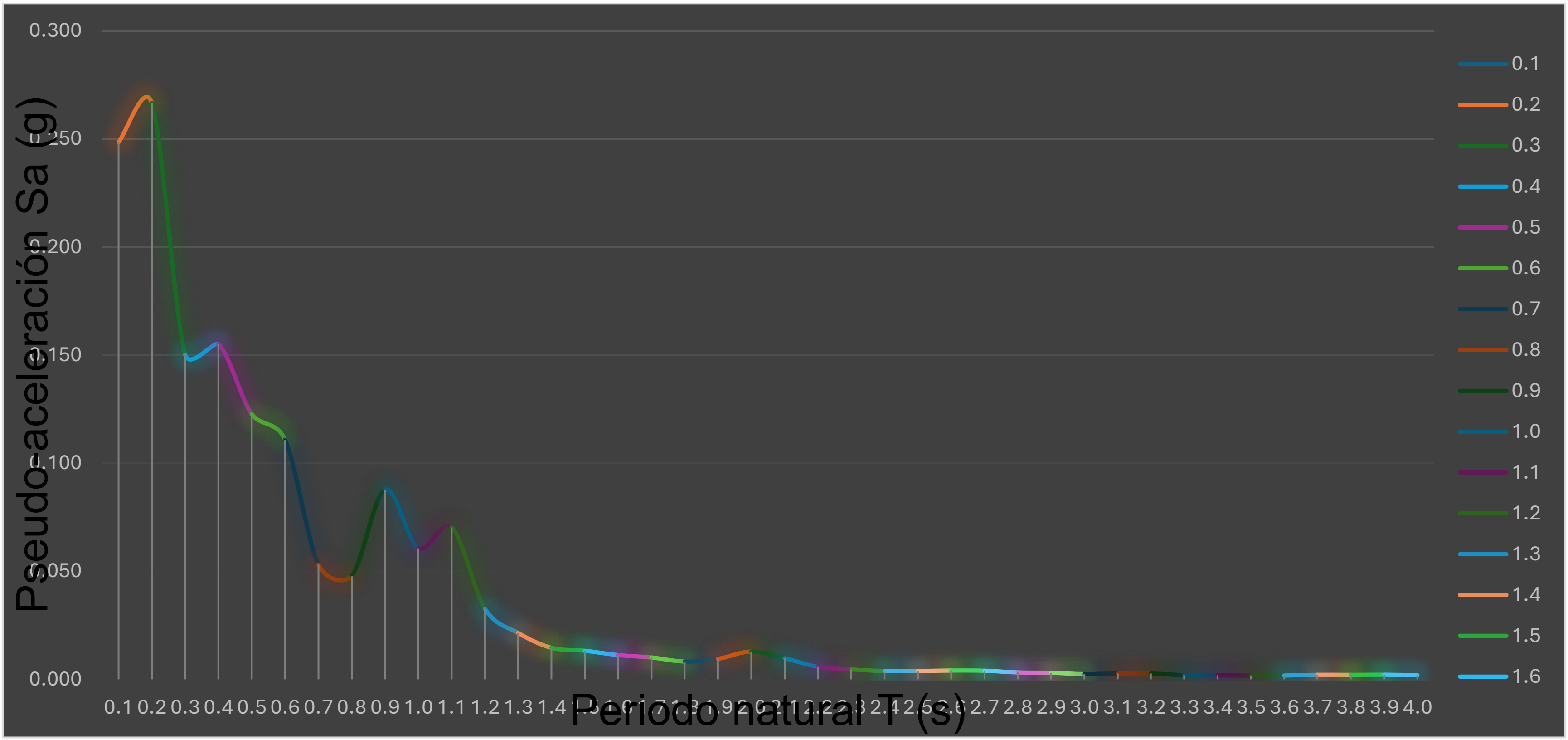
VALIDACIÓN DEL MÉTODO – T = 0.1 s, ξ = 5%, t* = 0.005 s

T (s)	0.100000
ξ	0.050000
t* (s)	0.005000
ω_n (rad/s)	62.831853
ω_a (rad/s)	62.753264

n	t	ag (gal)	α	A	B	C	D	u(t) (cm)	$\dot{u}$ (cm/s)	$\ddot{u}(t)$ (cm/s^2)
1	0.000	-0.006490	-1.298000	0.000001	-0.000329	-0.000001	0.000005	0.000000	-0.000016	-0.006285
2	0.005	-0.033550	-5.412000	0.000001	-0.001371	-0.000001	0.000022	0.000000	-0.000112	-0.031688
3	0.010	-0.055030	-4.296000	-0.000007	-0.001088	0.000006	0.000016	-0.000001	-0.000313	-0.047849
4	0.015	0.042420	19.490000	-0.000022	0.004937	0.000020	-0.000083	-0.000003	-0.000290	0.056266
5	0.020	0.373440	66.204000	-0.000016	0.016770	0.000013	-0.000271	-0.000002	0.000809	0.377915
6	0.025	0.894550	104.222000	0.000053	0.026400	-0.000055	-0.000411	0.000008	0.003878	0.837269
7	0.030	1.398380	100.766000	0.000186	0.025524	-0.000178	-0.000354	0.000040	0.008980	1.185001
8	0.035	1.707420	61.808000	0.000329	0.015656	-0.000290	-0.000121	0.000100	0.015041	1.219161
9	0.040	1.866240	31.764000	0.000420	0.008046	-0.000320	0.000095	0.000189	0.020605	0.989316
10	0.045	2.136760	54.104000	0.000451	0.013705	-0.000262	0.000097	0.000304	0.025061	0.779407
11	0.050	2.803510	133.350000	0.000487	0.033778	-0.000184	-0.000148	0.000440	0.029257	0.884569
12	0.055	3.957630	230.824000	0.000617	0.058468	-0.000178	-0.000474	0.000599	0.034954	1.373177
13	0.060	5.403850	289.244000	0.000886	0.073266	-0.000287	-0.000625	0.000794	0.043459	1.997199
14	0.065	6.713890	262.008000	0.001263	0.066367	-0.000469	-0.000389	0.001037	0.054238	2.277593
15	0.070	7.383950	134.012000	0.001647	0.033946	-0.000609	0.000293	0.001335	0.064283	1.710114
16	0.075	7.033770	-70.036000	0.001899	-0.017740	-0.000564	0.001279	0.001671	0.068590	0.007449
17	0.080	5.570230	-292.708000	0.001900	-0.074144	-0.000229	0.002263	0.002002	0.061712	-2.721387
18	0.085	3.241900	-465.666000	0.001599	-0.117955	0.000403	0.002883	0.002263	0.039834	-5.940991
19	0.090	0.542380	-539.904000	0.001039	-0.136759	0.001224	0.002875	0.002375	0.002516	-8.847954
20	0.095	-1.988670	-506.210000	0.000341	-0.128224	0.002033	0.002185	0.002268	-0.046656	-10.649239

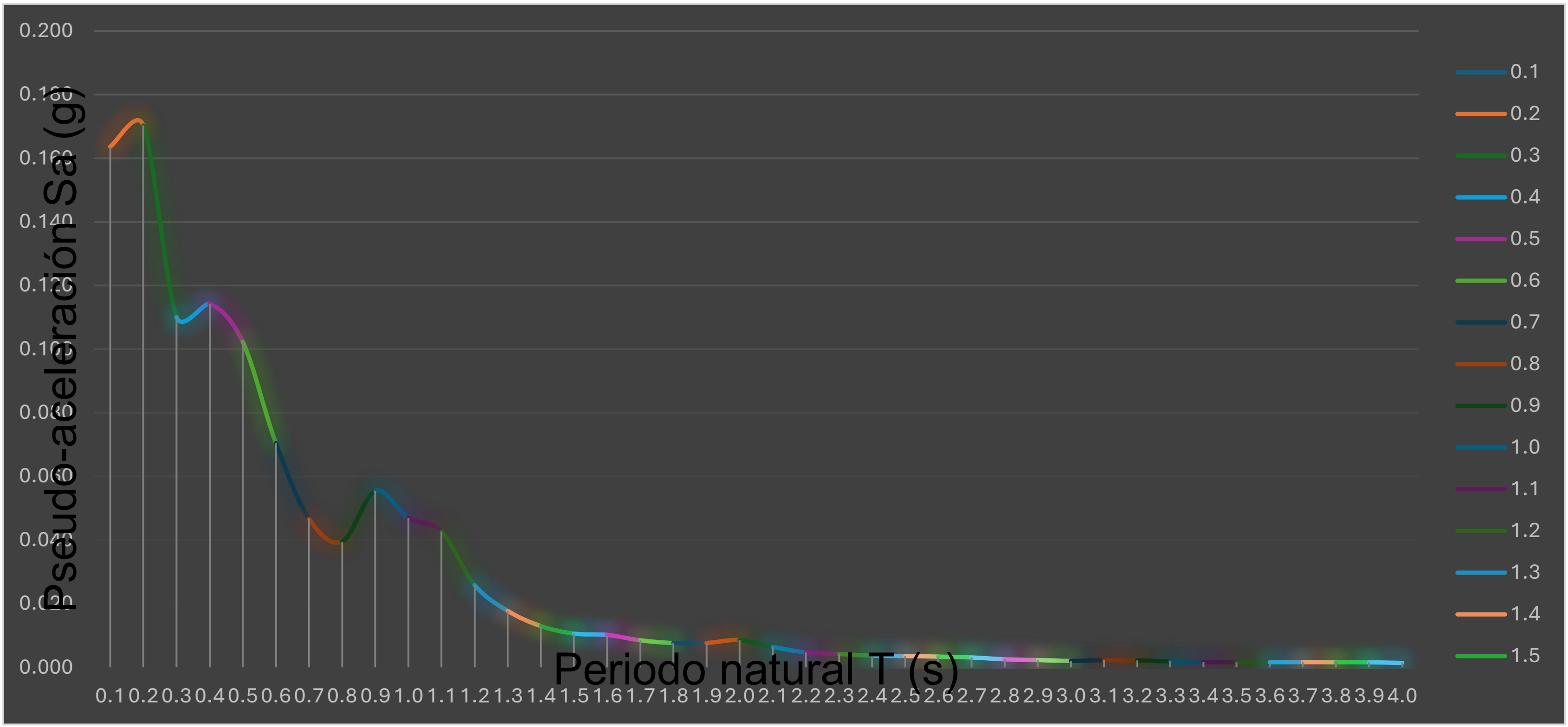
ESPECTRO ELÁSTICO LINEAL – XN003 – AMORTIGUAMIENTO ξ = 2%						
Period	**ξ**	**ω_n (rad/s)**	**Sd máx (cm)**	**Sv pseudo**	**Sa pseudo**	**Sa pseudo**
0.1	2%	62.8319	0.061751	3.8799	243.7831	0.248505
0.2	2%	31.4159	0.264963	8.3241	261.5079	0.266573
0.3	2%	20.9440	0.336130	7.0399	147.4432	0.150299
0.4	2%	15.7080	0.616432	9.6829	152.0986	0.155044
0.5	2%	12.5664	0.761949	9.5749	120.3222	0.122653
0.6	2%	10.4720	0.991470	10.3826	108.7268	0.110833
0.7	2%	8.9760	0.646918	5.8067	52.1210	0.053131
0.8	2%	7.8540	0.761827	5.9834	46.9933	0.047904
0.9	2%	6.9813	1.766641	12.3335	86.1039	0.087772
1.0	2%	6.2832	1.505354	9.4584	59.4290	0.060580
1.1	2%	5.7120	2.115458	12.0835	69.0206	0.070357
1.2	2%	5.2360	1.170991	6.1313	32.1034	0.032725
1.3	2%	4.8332	0.901079	4.3551	21.0492	0.021457
1.4	2%	4.4880	0.711110	3.1915	14.3232	0.014601
1.5	2%	4.1888	0.740517	3.1019	12.9931	0.013245
1.6	2%	3.9270	0.716033	2.8119	11.0421	0.011256
1.7	2%	3.6960	0.727645	2.6894	9.9399	0.010132
1.8	2%	3.4907	0.666060	2.3250	8.1157	0.008273
1.9	2%	3.3069	0.854004	2.8241	9.3393	0.009520
2.0	2%	3.1416	1.281439	4.0258	12.6473	0.012892
2.1	2%	2.9920	1.072555	3.2091	9.6015	0.009788
2.2	2%	2.8560	0.698804	1.9958	5.6999	0.005810
2.3	2%	2.7318	0.606738	1.6575	4.5280	0.004616
2.4	2%	2.6180	0.554086	1.4506	3.7976	0.003871
2.5	2%	2.5133	0.600702	1.5097	3.7944	0.003868
2.6	2%	2.4166	0.688751	1.6644	4.0223	0.004100
2.7	2%	2.3271	0.725594	1.6885	3.9294	0.004005
2.8	2%	2.2440	0.627644	1.4084	3.1605	0.003222
2.9	2%	2.1666	0.635174	1.3762	2.9816	0.003039
3.0	2%	2.0944	0.549995	1.1519	2.4125	0.002459
3.1	2%	2.0268	0.655879	1.3294	2.6944	0.002747
3.2	2%	1.9635	0.690175	1.3552	2.6608	0.002712
3.3	2%	1.9040	0.535118	1.0189	1.9399	0.001977
3.4	2%	1.8480	0.538752	0.9956	1.8399	0.001876
3.5	2%	1.7952	0.533919	0.9585	1.7207	0.001754
3.6	2%	1.7453	0.576962	1.0070	1.7575	0.001792
3.7	2%	1.6982	0.722213	1.2264	2.0827	0.002123
3.8	2%	1.6535	0.750067	1.2402	2.0507	0.002090
3.9	2%	1.6111	0.815648	1.3141	2.1171	0.002158
4.0	2%	1.5708	0.763189	1.1988	1.8831	0.001920

Indicador	Resultado
Sa máx (g)	0.266573
Periodo en Sa máx (s)	0.200000
PGA (g)	0.076958



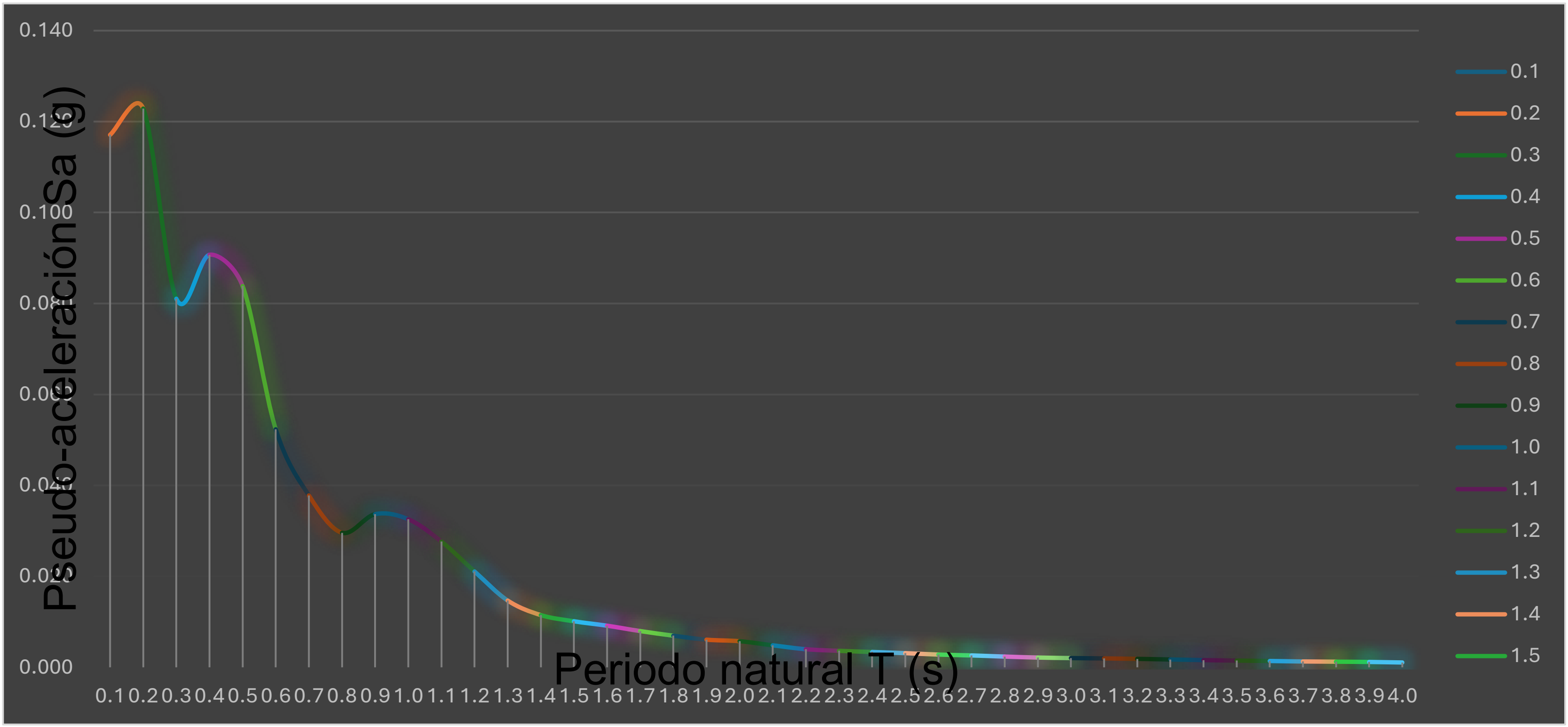
ESPECTRO ELÁSTICO LINEAL – XN003 – AMORTIGUAMIENTO ξ = 5%						
Periodo T (s)	ξ	ω_n (rad/s)	Sd máx (cm)	Sv pseudo (cm/s)	Sa pseudo (gal)	Sa pseudo (g)
0.1	5%	62.8319	0.040672	2.5555	160.5678	0.163678
0.2	5%	31.4159	0.169296	5.3186	167.0886	0.170325
0.3	5%	20.9440	0.246264	5.1577	108.0235	0.110116
0.4	5%	15.7080	0.454188	7.1344	112.0663	0.114237
0.5	5%	12.5664	0.635598	7.9872	100.3696	0.102314
0.6	5%	10.4720	0.630436	6.6019	69.1351	0.070474
0.7	5%	8.9760	0.568802	5.1056	45.8273	0.046715
0.8	5%	7.8540	0.628382	4.9353	38.7618	0.039512
0.9	5%	6.9813	1.115517	7.7878	54.3690	0.055422
1.0	5%	6.2832	1.166807	7.3313	46.0637	0.046956
1.1	5%	5.7120	1.281667	7.3209	41.8167	0.042627
1.2	5%	5.2360	0.924998	4.8433	25.3593	0.025851
1.3	5%	4.8332	0.745117	3.6013	17.4059	0.017743
1.4	5%	4.4880	0.631474	2.8340	12.7192	0.012966
1.5	5%	4.1888	0.592473	2.4817	10.3955	0.010597
1.6	5%	3.9270	0.648916	2.5483	10.0071	0.010201
1.7	5%	3.6960	0.609824	2.2539	8.3304	0.008492
1.8	5%	3.4907	0.615511	2.1485	7.4998	0.007645
1.9	5%	3.3069	0.688942	2.2783	7.5342	0.007680
2.0	5%	3.1416	0.852306	2.6776	8.4119	0.008575
2.1	5%	2.9920	0.703014	2.1034	6.2934	0.006415
2.2	5%	2.8560	0.563889	1.6105	4.5995	0.004689
2.3	5%	2.7318	0.552317	1.5088	4.1219	0.004202
2.4	5%	2.6180	0.528493	1.3836	3.6222	0.003692
2.5	5%	2.5133	0.552016	1.3874	3.4868	0.003554
2.6	5%	2.4166	0.566997	1.3702	3.3113	0.003375
2.7	5%	2.3271	0.562428	1.3088	3.0458	0.003105
2.8	5%	2.2440	0.493391	1.1072	2.4845	0.002533
2.9	5%	2.1666	0.477309	1.0341	2.2406	0.002284
3.0	5%	2.0944	0.453029	0.9488	1.9872	0.002026
3.1	5%	2.0268	0.534849	1.0840	2.1972	0.002240
3.2	5%	1.9635	0.554850	1.0894	2.1391	0.002181
3.3	5%	1.9040	0.507098	0.9655	1.8383	0.001874
3.4	5%	1.8480	0.480739	0.8884	1.6418	0.001674
3.5	5%	1.7952	0.478460	0.8589	1.5419	0.001572
3.6	5%	1.7453	0.515368	0.8995	1.5699	0.001600
3.7	5%	1.6982	0.553266	0.9395	1.5955	0.001626
3.8	5%	1.6535	0.576090	0.9525	1.5750	0.001606
3.9	5%	1.6111	0.599284	0.9655	1.5555	0.001586
4.0	5%	1.5708	0.555108	0.8720	1.3697	0.001396

Indicador	Resultado
Sa máx (g)	0.170325
Periodo en Sa máx (s)	0.200000
PGA (g)	0.076958

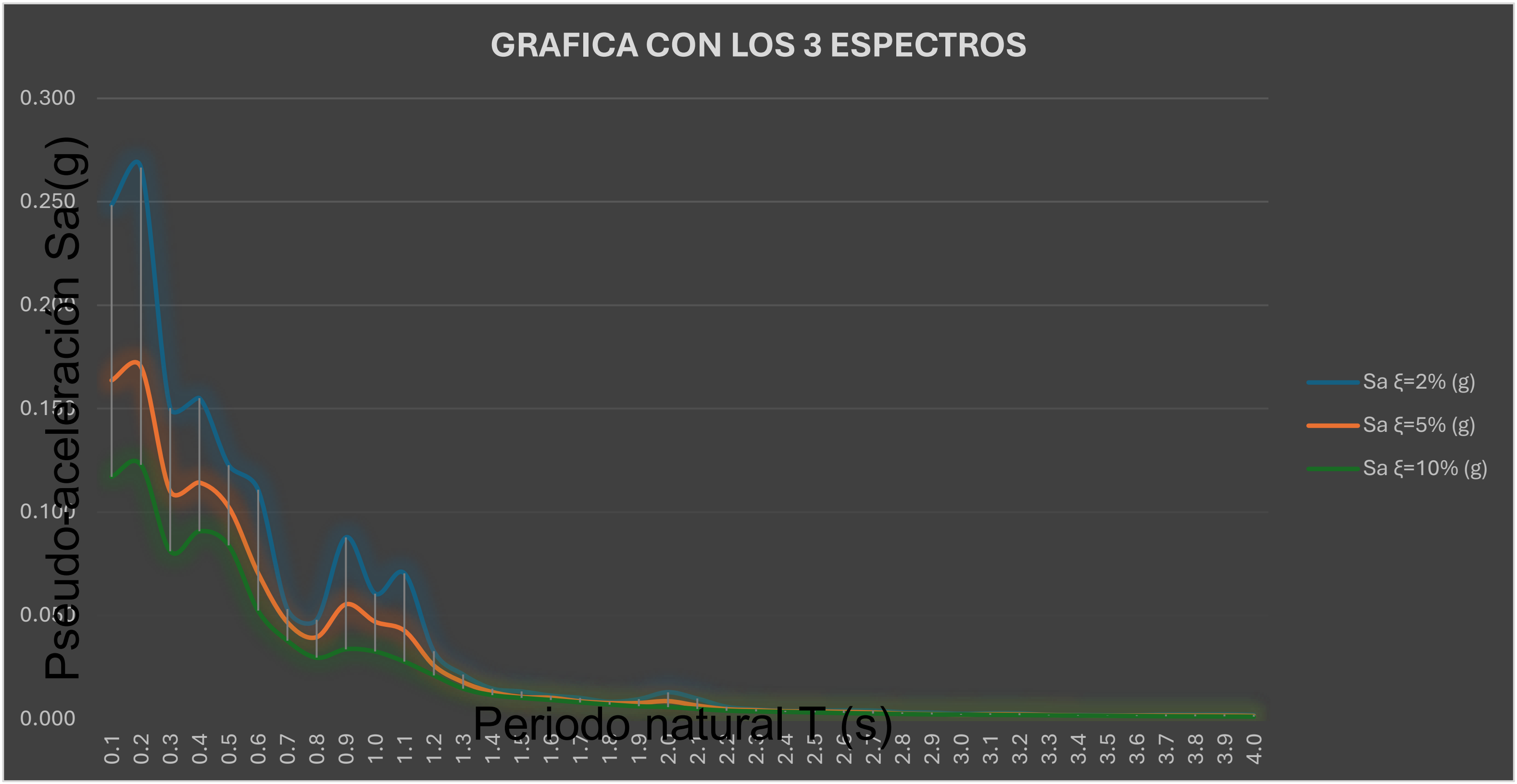


ESPECTRO ELÁSTICO LINEAL – XN003 – AMORTIGUAMIENTO ξ = 10%						
Periodo T (s)	ξ	ω_n (rad/s)	Sd máx (cm)	Sv pseudo (cm/s)	Sa pseudo (gal)	Sa pseudo (g)
0.1	10%	62.8319	0.029097	1.8282	114.8700	0.117095
0.2	10%	31.4159	0.122117	3.8364	120.5248	0.122859
0.3	10%	20.9440	0.181266	3.7964	79.5123	0.081052
0.4	10%	15.7080	0.360654	5.6651	88.9877	0.090711
0.5	10%	12.5664	0.520748	6.5439	82.2332	0.083826
0.6	10%	10.4720	0.468217	4.9032	51.3457	0.052340
0.7	10%	8.9760	0.460229	4.1310	37.0798	0.037798
0.8	10%	7.8540	0.470480	3.6951	29.0216	0.029584
0.9	10%	6.9813	0.677955	4.7330	33.0427	0.033683
1.0	10%	6.2832	0.809190	5.0843	31.9455	0.032564
1.1	10%	5.7120	0.831266	4.7482	27.1216	0.027647
1.2	10%	5.2360	0.754246	3.9492	20.6781	0.021079
1.3	10%	4.8332	0.616552	2.9799	14.4027	0.014682
1.4	10%	4.4880	0.559423	2.5107	11.2679	0.011486
1.5	10%	4.1888	0.565963	2.3707	9.9304	0.010123
1.6	10%	3.9270	0.582173	2.2862	8.9778	0.009152
1.7	10%	3.6960	0.572516	2.1160	7.8208	0.007972
1.8	10%	3.4907	0.560076	1.9550	6.8244	0.006957
1.9	10%	3.3069	0.545212	1.8030	5.9624	0.006078
2.0	10%	3.1416	0.570955	1.7937	5.6351	0.005744
2.1	10%	2.9920	0.530447	1.5871	4.7486	0.004841
2.2	10%	2.8560	0.478491	1.3666	3.9029	0.003978
2.3	10%	2.7318	0.477219	1.3037	3.5614	0.003630
2.4	10%	2.6180	0.483276	1.2652	3.3123	0.003376
2.5	10%	2.5133	0.483814	1.2160	3.0560	0.003115
2.6	10%	2.4166	0.470805	1.1378	2.7495	0.002803
2.7	10%	2.3271	0.473582	1.1021	2.5646	0.002614
2.8	10%	2.2440	0.460253	1.0328	2.3176	0.002362
2.9	10%	2.1666	0.449130	0.9731	2.1083	0.002149
3.0	10%	2.0944	0.454941	0.9528	1.9956	0.002034
3.1	10%	2.0268	0.468523	0.9496	1.9247	0.001962
3.2	10%	1.9635	0.473576	0.9299	1.8258	0.001861
3.3	10%	1.9040	0.466550	0.8883	1.6913	0.001724
3.4	10%	1.8480	0.456633	0.8439	1.5594	0.001590
3.5	10%	1.7952	0.451569	0.8107	1.4553	0.001483
3.6	10%	1.7453	0.449155	0.7839	1.3682	0.001395
3.7	10%	1.6982	0.439125	0.7457	1.2663	0.001291
3.8	10%	1.6535	0.446155	0.7377	1.2198	0.001243
3.9	10%	1.6111	0.445476	0.7177	1.1563	0.001179
4.0	10%	1.5708	0.423839	0.6658	1.0458	0.001066

Indicador	Resultado
Sa máx (g)	0.122859
Periodo en Sa máx (s)	0.200000
PGA (g)	0.076958



COMPARACIÓN DE LOS TRES ESPECTROS ELÁSTICOS – XN003			
Periodo T (s)	Sa ξ=2% (g)	Sa ξ=5% (g)	Sa ξ=10% (g)
0.1	0.248505	0.163678	0.117095
0.2	0.266573	0.170325	0.122859
0.3	0.150299	0.110116	0.081052
0.4	0.155044	0.114237	0.090711
0.5	0.122653	0.102314	0.083826
0.6	0.110833	0.070474	0.052340
0.7	0.053131	0.046715	0.037798
0.8	0.047904	0.039512	0.029584
0.9	0.087772	0.055422	0.033683
1.0	0.060580	0.046956	0.032564
1.1	0.070357	0.042627	0.027647
1.2	0.032725	0.025851	0.021079
1.3	0.021457	0.017743	0.014682
1.4	0.014601	0.012966	0.011486
1.5	0.013245	0.010597	0.010123
1.6	0.011256	0.010201	0.009152
1.7	0.010132	0.008492	0.007972
1.8	0.008273	0.007645	0.006957
1.9	0.009520	0.007680	0.006078
2.0	0.012892	0.008575	0.005744
2.1	0.009788	0.006415	0.004841
2.2	0.005810	0.004689	0.003978
2.3	0.004616	0.004202	0.003630
2.4	0.003871	0.003692	0.003376
2.5	0.003868	0.003554	0.003115
2.6	0.004100	0.003375	0.002803
2.7	0.004005	0.003105	0.002614
2.8	0.003222	0.002533	0.002362
2.9	0.003039	0.002284	0.002149
3.0	0.002459	0.002026	0.002034
3.1	0.002747	0.002240	0.001962
3.2	0.002712	0.002181	0.001861
3.3	0.001977	0.001874	0.001724
3.4	0.001876	0.001674	0.001590
3.5	0.001754	0.001572	0.001483
3.6	0.001792	0.001600	0.001395
3.7	0.002123	0.001626	0.001291
3.8	0.002090	0.001606	0.001243
3.9	0.002158	0.001586	0.001179
4.0	0.001920	0.001396	0.001066



Interpretación: para el registro XN003, la respuesta espectral disminuye al aumentar el amortiguamiento. La mayor pseudo-aceleración se presenta alrededor de T = 0.2 s en los tres casos.